SSRGD: Simple Stochastic Recursive Gradient Descent for Escaping Saddle Points

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The paper in one minute

Question: Can a near-optimal first-order variance-reduced algorithm gain second-order saddle-escape guarantees simply by adding occasional random perturbations?
Model/Method: SSRGD uses a SPIDER/SARAH-like recursive gradient estimator, snapshot resets, and perturbation-triggered "super epochs" when the gradient is small.
Finding: For finite sums it achieves $O(n + \sqrt{n}/\varepsilon^2)$ first-order gradients and $\widetilde{O}(\sqrt{n}/\varepsilon^2 + \sqrt{n}/\delta^4 + n/\delta^3)$ for (ε, δ) second-order stationarity, with relevant constants suppressed.
Why we care: A simple, well-controlled stochastic recursion plus perturbation can substitute for a dedicated negative-curvature detector.

1. Start with the recursive gradient estimator

For $f(x) = \frac{1}{n}\sum_{i=1}^{n} f_i(x)$, SGD's sampled gradients can vary greatly between examples. SSRGD instead starts each epoch with a full gradient and corrects a running estimate via matched component-gradient differences:

$$v_t = v_{t-1} + \frac{1}{b}\sum_{i \in I_b} (\nabla f_i(x_t) - \nabla f_i(x_{t-1})), \quad x_{t+1} = x_t - \eta v_t.$$

The same sampled indices appear at the new and previous point, which often cancels much of the sampling variability. The baseline is refreshed periodically to avoid accumulating uncontrolled error.

Worked example: why the recursive correction can be accurate

Let $f_1(x) = \frac{1}{2}(x-1)^2$ and $f_2(x) = \frac{1}{2}(x+1)^2$. Thus $f'(x) = x$. Take $x_0 = 2$ and full-gradient estimate $v_0 = 2$. A step of length 0.1 gives $x_1 = 1.8$. Whichever single component is sampled, its gradient difference is

$$\nabla f_i(1.8) - \nabla f_i(2) = -0.2.$$

So $v_1 = 2 - 0.2 = 1.8 = f'(1.8)$, despite sample gradients 0.8 and 2.8 individually being noisy. This toy example is not a general unbiasedness claim for the recursion.

First-order versus second-order: The recursive estimator reduces gradient noise efficiently. By itself it only supports fast search for small gradients. To escape saddles, SSRGD adds a separate *random perturbation* rule.

2. How a simple perturbation handles saddles

Algorithm 1: trigger a super epoch

At an epoch start, if the full gradient norm falls below threshold g_{thres} and the method is not already in a super epoch, draw ξ uniformly from a radius-r ball and set $x \leftarrow x + \xi$. Run the same recursive gradient updates, but avoid immediately perturbing again. A super-epoch stopping rule tracks escape over a controlled time window.

A two-dimensional escape calculation

Take $f(x, y) = \frac{1}{2}x^2 - \frac{1}{2}y^2$, a strict saddle at the origin. Ordinary gradient descent at $(0, 0)$ is motionless. Perturb to $(0.01, 0.01)$ and use $\eta = 0.1$:

$$x_{t+1} = 0.9x_t, \qquad y_{t+1} = 1.1y_t.$$

After 10 idealized full-gradient steps, $x_{10} \approx 0.00349$ while $y_{10} \approx 0.02594$. The negative-curvature direction becomes dominant. SSRGD's actual stochastic trajectory needs a nontrivial analysis to ensure estimator error does not overwhelm this growth.

Theorem 1: first-order rate

With $m = b = \sqrt{n}$ and a suitably small step size, the finite-sum method reaches an ε-first-order stationary point in expectation using

$$O\left(n + \frac{\sqrt{n}}{\varepsilon^2}\right)$$

stochastic gradients, with the smoothness constant and initial objective gap suppressed.

Theorem 2: escape / second-order rate

Under the additional Hessian-Lipschitz assumption and calibrated perturbations, it reaches

$$\|\nabla f(x)\| \le \varepsilon, \quad \lambda_{\min}(\nabla^2 f(x)) \ge -\delta$$

with high probability using

$$\widetilde{O}\left(\frac{\sqrt{n}}{\varepsilon^2} + \frac{\sqrt{n}}{\delta^4} + \frac{n}{\delta^3}\right)$$

finite-sum stochastic gradients (again suppressing problem-dependent constants). Theorem 3 and later results separately treat online expectation objectives with their own variance assumptions.

Read: Algorithm 1 and Table 1, Section 4's Theorems 1–2, and Section 5's proof overview. Compare the online guarantees in Tables 1–2 before transferring conclusions across oracle settings.

3. Results, assumptions, limitations, and relationships

Contributions. A recursive-gradient algorithm with near-optimal finite-sum first-order complexity; a simple uniform-perturbation modification yielding approximate second-order stationarity; online extensions; a simpler route to strong bounds than separate NEON2 modules in the regimes compared. **Assumptions.** Lower-bounded smooth objective, component-gradient Lipschitz continuity, Hessian Lipschitz continuity for saddle-escape theorem, bounded variance for online results, suitable batch size b, epoch length m, step size, perturbation radius and stopping thresholds. **Limitations.** Full gradients at snapshot epochs can be expensive when n is huge; the complexity bound concerns stochastic gradient evaluations rather than total elapsed time. The guarantee is approximate second-order stationarity, not global optimality; the perturbation schedule must satisfy the theorem's conditions.

Cross-paper comparison

Stabilized SVRG (Ref 82) uses an explicit gradient *shift* inside super epochs; SSRGD instead uses recursive SPIDER/SARAH gradient differences and a perturbation. **SpiderBoost (Ref 80)** has a related recursive estimator but its main theorems are first-order. **NEON2 (Ref 77)** employs a separate negative-curvature finder; SSRGD directly modifies its existing optimizer. **Jin (2017)** motivates why a random kick can prevent stagnation at strict saddles.

Quick reference

Concept	Interpretation
Recursive v_t	Correct prior estimate using matched sample-gradient differences.
Super epoch	Perturb only occasionally near a low-gradient candidate.
$O(n + \sqrt{n}/\varepsilon^2)$	Finite-sum first-order stochastic-gradient complexity order.
(ε, δ) point	Small gradient, smallest Hessian eigenvalue at least $-\delta$.
$\widetilde{O}(\sqrt{n}/\varepsilon^2 + \sqrt{n}/\delta^4 + n/\delta^3)$	Second-order finite-sum complexity order (constants hidden).

Papers Master note

Li's SSRGD combines a recursive SPIDER/SARAH-style estimator with occasional uniform perturbations, attaining near-optimal first-order finite-sum complexity and a second-order saddle-escape guarantee without a separate eigenvector search. Finite-sum/online rates require different assumptions.

Verified source: original NeurIPS 2019 PDF, Algorithm 1, Theorems 1–3, Section 5, Tables 1–2.